

Design and Manufacture of a CFRP Floor Cover for the E16 Airfield Retrieval Vehicle

A composites design-and-build project – from reverse engineering through mould making to the finished laminate part.

Context	Aviation Engineering Association (FTAG), Esslingen University of Applied Sciences
Project	E16 – electric airfield retrieval vehicle
Role	Design, mould making and lamination (fully independent)
Methods & tools	3D scanning (Creality / CrealityScan), reverse engineering & CAD (Fusion 360), FDM printing, mould making, hand lamination with vacuum bagging
Materials	CFRP (4 plies, 0/45/0/45 orientation), L285 epoxy resin, approx. 3 kg PLA, wood, press-fit threaded inserts

Background and Objective

The E16 is the 16th project of the Aviation Engineering Association (FTAG) at Esslingen University of Applied Sciences. It is an electric airfield retrieval vehicle built on the chassis of a quad bike and the fuselage of a two-seat glider, driven by a 10 kW electric powertrain.

The vehicle's brake cylinder sits in the nose of the fuselage, behind the pedals. From there, the brake lines run across the floor of the fuselage to the front and rear wheels. When entering and exiting the vehicle, there was a risk of stepping on these unprotected lines and damaging them.

To prevent this, a load-bearing floor cover had to be designed and manufactured for the driver's area – a part that did not previously exist. This report describes the process from the initial survey through to the installed component.

Preparation: 3D Scan of the Footwell

The brake lines are not represented in the vehicle's existing CAD model, as they were routed by hand. In addition, several other parts had been bonded in place without being added to the CAD model. This resulted in a significant discrepancy between the digital model and the actual vehicle.

To account for these manufacturing-related deviations during design and to avoid overlooking any relevant components, the driver's footwell was captured using a Creality 3D scanner. The resulting scan served as the reference geometry for all subsequent design work.

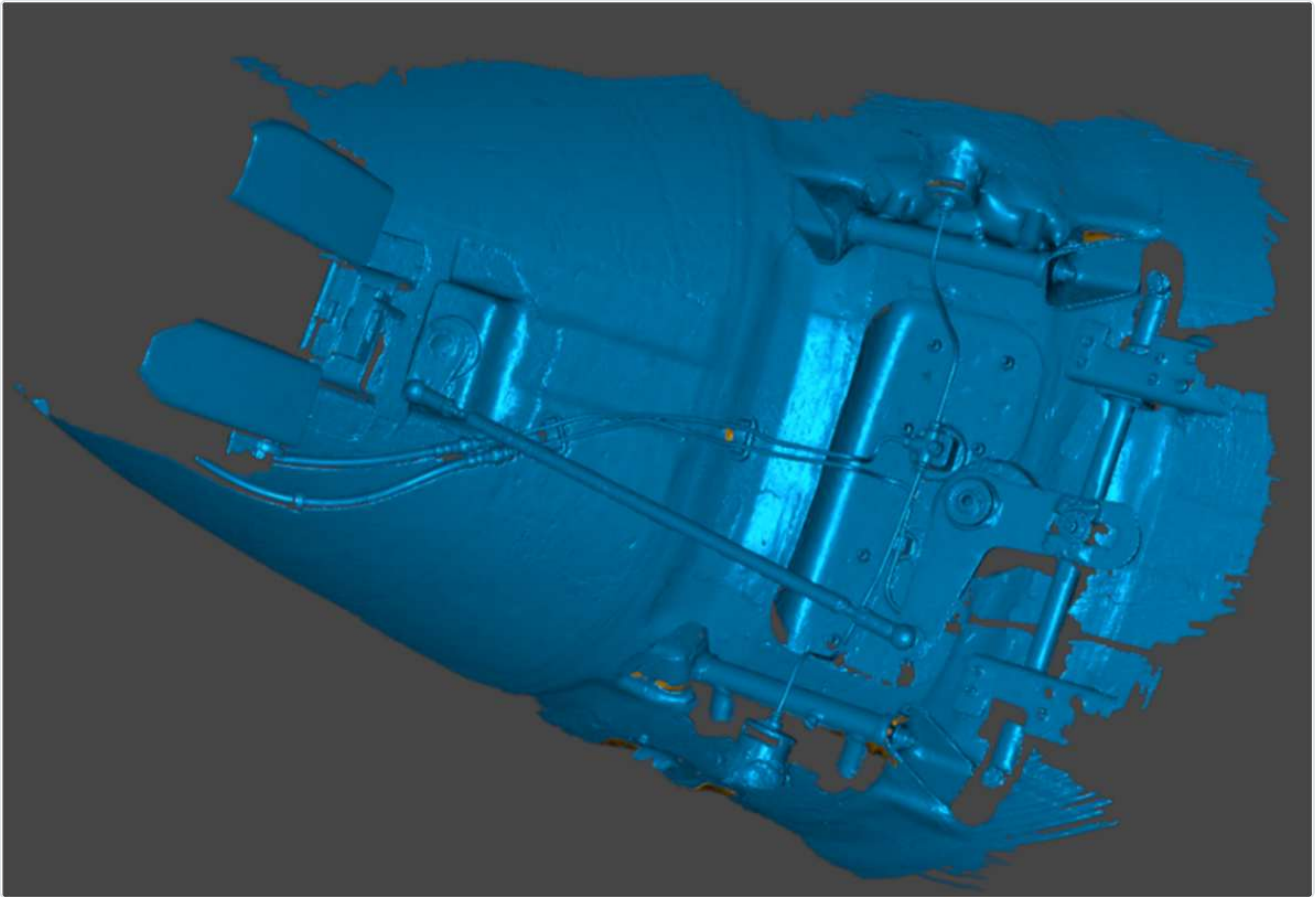


Figure 1: 3D scan of the front fuselage area in CrealtyScan

Reverse Engineering and Design

The 3D scan was first cleaned up in CrealtyScan – small holes were closed and the triangle count of the mesh was reduced – and then exported to Fusion 360. The work in Fusion 360 proceeded in several steps:

- **Reconstructed the fuselage geometry as a surface:** The fuselage contour was traced with splines on planes spaced 30 mm apart and joined into a continuous surface using the *Loft* tool.
- **Re-modelled surrounding components:** Key parts such as the steering column and the pass-through fittings for the brake line were added as solid bodies to rule out collisions.
- **Designed the mounting blocks:** Also using the *Loft* tool, mounting blocks were created to match the fuselage form as closely as possible. These were to be FDM-printed and fitted with press-fit threaded inserts so that the floor cover would remain removable for maintenance.
- **Derived the floor cover:** Once the position and geometry of the mounting blocks were fixed and represented in the CAD model, the cover itself was designed as a surface. For manufacturing reasons, it was split into a front and a rear section.

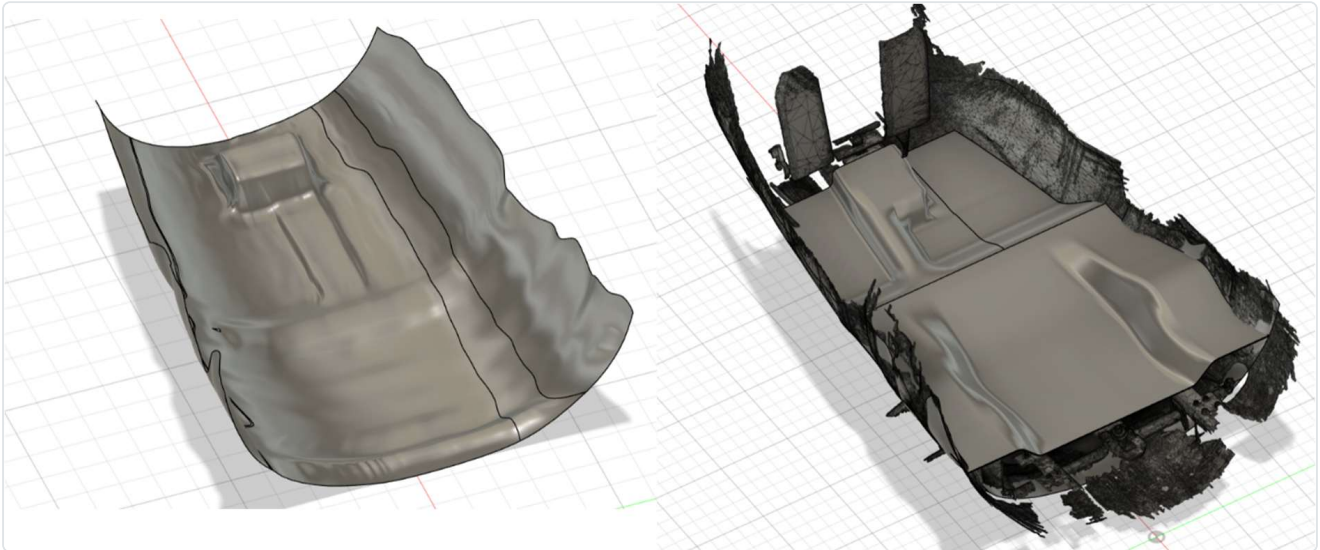


Figure 2: Left, the reconstructed fuselage shell; right, the CAD model of the floor cover overlaid on the scan

Mould Making and Lamination

The decision was made to manufacture the floor cover from CFRP. To be able to laminate the part, a mould was derived from the CAD model. This mould was divided into several smaller segments and FDM-printed; in total, around 3 kg of PLA filament was used.

Note on the manufacturing method: It would have been preferable to mill the mould from polymer foam. However, this manufacturing option was not available and would likely have been more expensive – though more durable.

During mould making, the concept was adapted: the cover was given additional angled side sections that are also laminated. To achieve this, the moulds were extended with bolt-on side pieces, allowing the curvature of the fuselage to be taken into account. To prevent the cover from flexing, additional mounting blocks were made from wood.

The individual mould segments were bonded together and the surface was sanded to ensure a clean underside on the floor cover as well. (Wet-sanding PLA is tedious but works reliably.)

The floor cover was then laminated from **4 plies of CFRP in a 0/45/0/45 layup**. **L285 epoxy resin** was used as the matrix system, and curing was carried out under vacuum.



Figure 3: The finished moulds – left, the rear mould during wet-sanding; right, the front mould after sanding



Figure 4: Rear mould under vacuum after lamination

In parallel with laminating the moulds, the angled side sections were laminated and the wooden mounting blocks were fabricated.

Result

The finished, multi-part CFRP component was post-processed and fitted into the vehicle's footwell. The brake lines are now protected, and thanks to the threaded inserts the cover remains removable for maintenance.



Figure 5: The finished parts installed in the vehicle

Conclusion and Learnings

This project let me apply several methods hands-on for the first time: 3D scanning, the reverse engineering built on it, and mould making for larger laminated components. I also had little prior experience working with fibre-reinforced plastics and learned a great deal here.

In hindsight, I significantly underestimated the effort involved – a lot of time and work ultimately went into the part. This was mainly down to a lack of experience, and that is precisely where the learning came from.

The most important takeaway: **small mistakes in design and planning quickly turn into major problems in manufacturing.** Accordingly, a substantial amount of rework was needed to fit the parts cleanly into the footwell. For future projects, I am taking with me just how valuable careful tolerance and fit considerations are already in the early design phase.